B.M.S COLLEGEOF ENGINEERING, BANGALORE-19
(Autonomous Institute, Affiliated to VTU)
Department of Computer Science and Engineering

INTERNALS -II

Course Code: 23CS4ESCRP	**Course Title :** Cryptography	
Semester: IV	**Maximum Marks: 40**	**Date:28/06/2024**
Faculty Handling the Course:	Dr. NV, Prof. SA, Dr. MDR, Prof. PP	

Instructions: *No choice in Part A and Part B. Internal choice is provided in Part C.*

PART-A

No.	Question	Marks	CO	BL
1.	Compare and contrast stream ciphers and block ciphers in terms of their structure, operation, strengths, and weaknesses with examples. **Solution: Each carries 1 Marks**	5	1	2

Criteria	**Stream Ciphers**	**Block Ciphers**
Structure	Encrypts data one bit or byte at a time.	Encrypts data in fixed-size blocks (e.g., 64 or 128 bits).
Operation	Typically uses a keystream generator and XOR operation with plaintext.	Divides plaintext into blocks and processes each block separately.
Key	Uses a key to generate a keystream that is combined with plaintext.	Uses a key to perform multiple rounds of complex operations on blocks.
Strengths	- Fast and suitable for real-time applications - Simple implementation	- High diffusion and resistance to cryptanalysis - Can use various modes of operation for security
Weaknesses	- Less secure if keystream is reused - Susceptible to bit-flipping attacks	- More complex and slower - Requires padding if the message size is not a multiple of the block size
Examples	RC4, A5/1	AES, DES, 3DES

PART-B

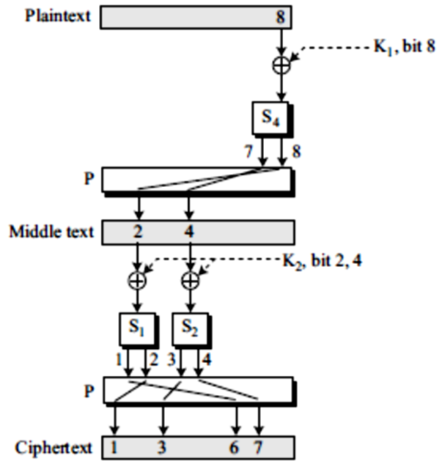
No.	Question	Marks	CO	BL
2.a	Imagine you are a cryptanalyst tasked with breaking a double DES encryption scheme used to secure sensitive communication between two parties. You have access to a plaintext-ciphertext pair (P,C). Devise a meet-in-the-middle attack to find the encryption keys K_1 and K_2. Justify why this attack is effective against double DES, and describe each step of the attack process in detail with neat diagram. **Meet-in-the-middle attack explanation - 4 marks** **Diagram- 1 marks**	5	2	4

2. b	Analyze the structural differences, strengths, weaknesses, and performance efficiency of the Advanced Encryption Standard (AES) compared to the Data Encryption Standard (DES), and explain why AES is preferred in modern cryptographic applications.	5	2	4

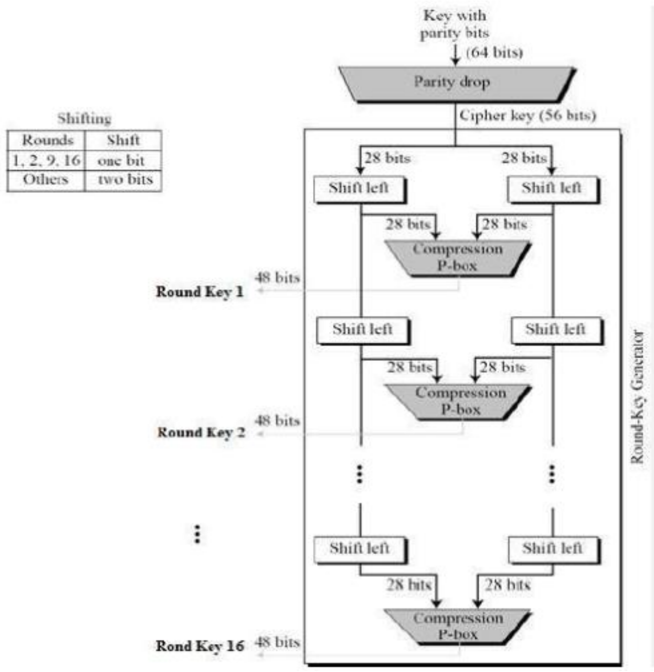
Each Carries 1 Marks

Criteria	Advanced Encryption Standard (AES)	Data Encryption Standard (DES)
Structure	- Symmetric block cipher - Block size: 128 bits - Key sizes: 128, 192, or 256 bits - Rounds: 10, 12, or 14 based on key size	- Symmetric block cipher - Block size: 64 bits - Key size: 56 bits - Rounds: 16
Strengths	- Stronger security due to larger key sizes - Resistant to various cryptographic attacks - Flexibility with different key lengths - Efficient implementation in hardware and software	- Simple and well-studied - Fast encryption/decryption process in hardware
Weaknesses	- More complex and potentially slower in software compared to DES	- Short key length makes it vulnerable to brute force attacks - Susceptible to certain cryptographic attacks (e.g., differential cryptanalysis)
Performance Efficiency	- Highly efficient in both hardware and software - Can handle larger blocks of data more efficiently - Designed for high throughput and low latency	- Efficient in hardware but slower in software - Inefficient with larger blocks of data - Limited by its smaller block and key sizes
Modern Preference	- AES is preferred due to its robust security and efficiency - Suitable for modern applications requiring high security - Widely adopted and standardized (e.g., by NIST)	- DES is considered obsolete for modern security needs - Mostly replaced by AES and 3DES (an extended version with higher security)

Reasons

2.c	Analyze how a product cipher achieves confusion and diffusion. With a neat diagram, illustrate the process. **Explanation : 3 Marks** **Diagram 2 Marks** **Figure 5.14** *Diffusion and confusion in a block cipher*  	5	2	4

PART- C

No.	Question -	Mark	CO	BL
3. a.	i. Demonstrate with a neat diagram key generation process in DES. Demonstration : 3 Marks Diagram : 2 Marks  ii. Write the pseudo code of SubBytes, ShiftRows, and MixColumns, for the AES encryption algorithm. **Subbyte : 1.5 Marks** **Shiftrows : 2 Marks** **Mix Columns : 1.5 Marks**	5 5	1	3

OR

3. b.	i. Calculate ϕ(210), ϕ(2310) using Euler's Totient Function. $210 = 2 \times 3 \times$ $\phi(210) = 2 \times 3 \times 5 \times 7$ $\phi(210)$ Factors are $2 \times 3 \times 5 \times 7$ $\phi(210) = \phi(2) \times \phi(3) \times \phi(5) \times \phi(7)$ 2.5 Marks $= 1 \times 2 \times 4 \times 6$ $= 48$ $\phi(2310)$ Factors are $2 \times 3 \times 5 \times 7 \times 11$ $= \phi(2) \times \phi(3) \times \phi(5) \times \phi(7) \times \phi(11)$ $= 1 \times 2 \times 4 \times 6 \times 10$ 2.5 Marks $= 480$ ii. Use Fermat's Little Theorem to find the remainder when a) 5^{50} is divided by 19. b) 6^{34} mod 17 1) Using Fermat's little theorem $a^{p-1} \equiv 1 \bmod p$ $a^{p} \equiv a \bmod p$ a) $5^{50} \bmod 19$ $= 5^{19} \times 5^{19} \times 5^{12} \bmod 19$ $= 5 \times 5 \times 11 \bmod 19$ $= 9.$ b) $6^{34} \bmod 17$ $= 6^{17} \times 6^{17} \bmod 17$ $= 6 \times 6 \bmod 17$ $= 2$	**5** **5**	**1**	**3**

4. a.	i. Solve the system of congruences using the Chinese Remainder Theorem and find the value of x.	**6**	**1**	**4**

$$x \equiv 2 \bmod 4$$
$$x \equiv 3 \bmod 7$$
$$x \equiv 1 \bmod 9.$$

4

Scheme

1 marks ↓   2 Marks ↓   1 Mark

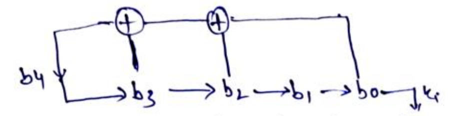
$a_1 = 2$	$m_1 = 4$	$M_1 = 63$	$M_1^{-1} = 63$	$M = 252$
$a_2 = 3$	$m_2 = 7$	$M_2 = 36$	$M_2^{-1} = 1$	
$a_3 = 1$	$m_3 = 9$	$M_3 = 28$	$M_3^{-1} = 1$	

3 marks

$$x = ((a_1 \times M_1 \times M_1^{-1}) + (a_2 \times M_2 \times M_2^{-1}) + (a_3 \times M_3 \times M_3^{-1})) \bmod M$$
$$= ((2 \times 63 \times 63) + (3 \times 36 \times 1) + (1 \times 28 \times 1)) \bmod 252$$
$$= (7938 + 108 + 28) \bmod 252 = 8074 \bmod 252$$
$$= 10$$

ii. Design a linear feedback shift registers with 4 cells in which $b_4 = b_3 \oplus b_2 \oplus b_0$. The initial state of the LFSR is [1,0,0,1], determine the next 8 states of the LFSR.

LFSR for $b_4 = b_3 \oplus b_2 \oplus b_0$



1 Mark

	b_4	b_3	b_2	b_1	b_0	k_r
Initial state	0	1	0	0	1	
1st step	1	0	1	0	0	1
2	1	1	0	1	0	0
3	1	1	1	0	1	0
4	0	1	1	1	0	1
5	0	0	1	1	1	0
6	1	0	0	1	1	1
7	0	1	0	0	1	1
8	1	0	1	0	0	1
						0

3 marks

Sequence 1 0 0 1 0 1 1 1 0
$k_{i=1}$

OR

4. b.		i. Apply the Miller-Rabin test to check if 631 is a prime number using base a=2. 4 Millers-Rabin test 631 $n-1 = 630 = 2 \times 315$ $a=2$, $m=315$ $k=1$ For loop $T \rightarrow a^{m} \bmod n \rightarrow 2^{315} \bmod 631 \rightarrow 1$ probably prime Each Step Carries Marks 5 marks ii. Find the modular inverse of the following using Euler's theorem a. 7 modulo 17 b. 5 modulo 19 Modular Inverse using Euler's theorem a) 7 modulo 17 According to Euler's theorem for finding Inverse $= a^{\phi(n)-1} \bmod n$ n is composite Since 17 is a prime, we have to apply Fermat's theorem $= a^{p-2} \bmod p$ $= 7^{17-2} \bmod 17 = 17^{15} \bmod 17$. $= 5$ 2) 5 modulo 19 Again Fermat's theorem $= 5^{19-2} \bmod 19$ $= 4$	**5** **5**	**1**	**3**